

Machine learning-based investigation of δ phase precipitation behavior in IN718 alloy and its influence on tensile properties

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ABSTRACT

Traditional experimental methods have limited the comprehensive understanding of how the δ phase affects the tensile properties of IN718 alloy. This study proposes a machine learning-based approach to systematically investigate the influence of the δ phase on tensile performance. The TernaNet-16 model was used in transfer learning on 1596 scanning electron microscope images, achieving Dice and IoU scores of 75.1 % and 80.6 %, respectively. It enabled automated extraction of the δ phase morphology and content from 36 groups of aged samples. Then a random forest regression model was developed to predict microstructural features and tensile properties with R^2 values exceeding 97 %. Shapley Additive Explanations (SHAP) analysis revealed the nuanced effects of the δ phase on tensile performance: (1) A small amount of δ phase at grain boundaries improves strength by hindering grain boundary motion, whereas excessive needle-like δ phase within and along grain boundaries enhances strength via the Orowan mechanism but reduces ductility. (2) The effect of the δ phase on tensile properties is nonlinear, influenced by its precipitation behavior and competition with the γ' phase. This work demonstrates a novel strategy for microstructure-performance correlation analysis and highlights the potential of machine learning in materials characterization and property prediction.

1. Introduction

IN718 alloy, also known as Inconel 718, is an age-hardenable nickel-based superalloy with a Ni-Cr-Fe matrix developed by the International Nickel Company (INCO). The numerical designation “718” refers specifically to its specific composition within the Inconel family. Due to its outstanding high-temperature strength up to approximately 650 °C, excellent corrosion and oxidation resistance, superior fatigue properties and fracture toughness, good machinability, and relatively low production costs, IN718 alloy is widely employed to manufacture critical components in aerospace engines and gas turbines, such as compressor disks, blades, casings, and turbine disks. It also finds extensive applications in automotive, marine, and energy sectors. It serves as the standard for research and development in advanced thermal processing technology and the design and management of high-heat-strength

structures in deformed high-temperature alloys [1]. The δ phase is the crucial precipitation phase in IN718 alloy; it can impede grain growth by pinning grain boundaries, hence influencing the hot working and mechanical properties of the material [2]. Meanwhile, excessive δ phase can facilitate crack initiation and propagation, resulting in diminished alloy plasticity and reduced mechanical performance of the material [3, 4]. Consequently, a thorough investigation of the influence of δ phase on IN718's mechanical properties is crucial for enhancing the alloy's characteristics.

So far, researchers have conducted comprehensive investigations into the precipitation behavior of δ phase in IN718 alloy and its influence on mechanical properties. These studies have mainly used various heat treatment schemes to modify the volume fraction, size, morphology, and distribution of δ phase, thereby facilitating a deeper understanding of its impact on IN718 alloy's mechanical characteristics.

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For example, Zhang et al. [5] conducted multi-pass cold rolling to produce tensile samples predominantly containing needle-like or rod-like δ phase, thereby evaluating the influence of δ phase on mechanical properties. Ye et al. [6] classified aging specimens into three categories based on the distribution, morphology, and content of δ phases: needle-like δ phase located at grain boundaries and twin boundaries (volume fractions from 2.20 % to 2.63 %); predominantly needle-like δ phase at grain and twin boundaries, with a minor presence of granular δ phase within the grain (volume fractions between 5.21 % and 6.09 %); and primarily granular phases distributed throughout the grain (volume fractions between 6.56 % and 7.56 %). The influence of morphology on mechanical properties was assessed by altering the δ phase content in samples from each group. Páramo-Kañetas et al. [7] initially assessed the effect of δ phase morphology on mechanical properties using a Delta Processing (DP) heat treatment method to induce a moderate precipitation of δ phase in IN718 alloy, followed by high-temperature compression deformation at 960 °C (below the dissolution temperature of δ phase) and 1020 °C (above the dissolution temperature of δ phase). Comparison of IN718 alloy samples subjected to different treatments revealed that the presence of δ phase enhanced the alloy's hardness and facilitated dynamic recrystallization during the hot compression process. However, current research primarily relies on limited compositional data and morphological observations. Most evaluations of δ phase's influence on material properties under various precipitation states are based on comparative experiments and lack precise or sufficient supporting data. To date, comprehensive and systematic investigations into the effect of δ phase on the mechanical properties of IN718 alloy remain limited.

The limitations of the current study largely arise from the deficiencies in the research technique. In the statistical analysis of the δ phase, researchers typically employ X-ray diffraction (XRD) to determine the overall content of the δ phase or utilize software like Image J for manual counting. These strategies either yield minimal information or require extensive manual effort. Therefore, there is an urgent need to develop a user-friendly and effective intelligent tool for δ phase statistics, which would provide strong support for future research in this field. The development and application of deep learning technologies have provided powerful research tools for the precise, rapid, and automated acquisition of information from material micrographs. Wang et al. [8] introduced an end-to-end convolutional neural network framework to achieve precise segmentation of the δ phase in IN718 alloy, demonstrating an absolute average error of only 1.968 in scanning electron microscopy images under a compressive strain of 0.1. Ma et al. [9] developed a deep convolutional neural network based on DeepLab for accurate segmentation of the microstructure of Al-La alloys while enhancing the resolution of microscopic images. Yoo et al. [10] applied a simple unsupervised machine learning method for downscaling and clustering with minimal hyperparameter tuning, enabling the semi-automatic identification of distinct coexisting structures in the alloy. Bilyk et al. [11] utilized a Mask Region-Based Convolutional Neural Network (Mask R-CNN) to quantitatively analyze the number, size, and spatial distribution of dislocation loops in ion-irradiated CrFeMnNi alloys based on transmission electron microscopy (TEM) images. Their work provided insights into the nucleation, motion, and growth mechanisms of dislocation loops, as well as the material's elastic anisotropy. Li et al. [12] proposed a method for uncertainty estimation using the Dirichlet distribution to facilitate the segmentation and measurement of aluminum alloy metallographic images. This approach effectively captured the uncertainty of classification distributions, improved segmentation accuracy and incorporated uncertainty estimation into the prediction process, thereby providing a more precise and reliable means of automated measurement for aluminum alloy microstructure. It can be seen that deep-learning based image recognition and segmentation methods demonstrate significant potential for characterizing material microstructures. It can enhance the efficiency of materials research and decrease costs, while also elucidating the intricate

relationships among material formation mechanisms, atomic structures, and material properties, thus serving as a powerful tool for materials exploration [13,14].

Machine learning facilitates the efficient processing and analysis of extensive datasets, extraction of essential features, high-throughput screening, and prediction of macroscopic properties, thereby reducing the time and cost associated with conventional trial-and-error approaches [15,16]. For example, Yu et al. [17] conducted feature extraction and statistical analysis on the microstructure of carbon nanotubes (CNTs) and employed the Extra Trees regression model to establish the structure-property relationship of the material, achieving prediction accuracy errors for hardness and density between 1.42 % and 2.47 %. Kusano et al. [18] utilized random forest regression to predict the mechanical properties of selective laser melting (SLM)-fabricated Ti6Al4V alloy based on its microstructural characteristics, achieving a prediction error below 2.0 %. Li et al. [19] employed three meta-models, including the Gaussian Process (GP), the Radial Basis Function (RBF), and the Support Vector Machine (SVM) to develop an integrated model for predicting the tensile strength of 316L stainless steel fabricated by Laser Powder Bed Fusion (L-PBF) using process data. E. Polyzos [20] introduced Maxwell's effective field method (EFM) that mitigates the inherent accuracy limitations of micromechanics-based models, while reducing the amount of data required for calibrating machine learning models.

With the continuous development of machine learning models, interpreting their decision-making process remains a significant challenge. These models often operate as "black boxes", making it difficult for researchers to understand how specific features influence predictions. This lack of interpretability limits their practical application, especially in fields such as materials science, where understanding underlying mechanisms is just as important as obtaining accurate predictions. Therefore, the development of interpretable algorithms and models is crucial for gaining deeper insights into the mapping between material data and machine learning models [21], which will accelerate the application of machine learning in materials knowledge mining and the understanding of material mechanisms. The core value of explainable machine learning lies in transforming "black-box" predictions into transparent mechanisms. This can be divided into two main categories: model-specific explanations and model-agnostic explanations. Model-specific explanations refer to explanations inherently provided by the model itself, such as feature weights in linear regression or split rules in decision trees [22]. However, model-specific explanations have a narrow range of applicability, usually limited to specific types of models, and often fail to provide clear explanations when handling complex nonlinear models. In contrast, model-agnostic explanation methods, such as SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), and Partial Dependence Plot (PDP), can explain any type of machine learning model, whether it is a simple linear regression model, a complex deep learning model, or ensemble method. PDP is a global method that shows how one or more features affect the model's prediction, but it cannot explain local feature effects [23]. LIME, a local explanation method, approximates a complex model's behavior by training a simple, interpretable model around a specific data point, but it lacks global consistency [24]. SHAP is based on Shapley values from game theory [25]. It decomposes a model's prediction by assigning each feature an additive contribution, ensuring fair and consistent feature importance evaluation. SHAP combines the strengths of PDP and LIME by providing explanations at both global and local levels. It reveals the driving factors behind the model's overall behavior and individual predictions, making it an ideal solution to address the "black-box" problem. For example, Wu et al. [26] employed the SHAP approach to conduct an interpretable study of a phase structure prediction model for high-entropy alloys, quantifying the contribution of components to the phase structure, which facilitated the optimization of alloy compositions. Jeon et al. [27] integrated SHAP with RFR model to examine the influence of tempering conditions on the

hardness of mild steel, concluding that tempering temperature, carbon content, and holding time are the three most significant factors affecting mild steel hardness. Xiong et al. [28] integrated the extremely random tree (ETR) algorithm with the Pearson correlation coefficient (PCC) and recursive feature elimination (RFE) methods to identify seven pivotal features influencing the entropy of isothermal magnetic transformation in amorphous alloys, elucidating the significance of these features and their critical values through the SHAP method. Liu et al. [29] effectively employed the RFR model to predict the Young's modulus of the alloy. They revealed the significant influence of key features such as average bulk modulus, average Young's modulus, and enthalpy of mixing of the alloy on the Young's modulus by SHAP analysis, which provides an important guidance for designing and optimizing new titanium alloys.

This study introduces a novel machine learning approach to systematically investigate the impact of the δ phase on the tensile properties of IN718 alloy. By employing a deep learning model to replace the traditional manual labeling of the δ phase, labor costs are significantly reduced, while providing a more precise and efficient method for optimizing material properties. Subsequently, the model was used to extract δ phase content and morphology as inputs and tensile properties as outputs. Coupled with SHAP analysis, an interpretable "structure-property" prediction model was developed, effectively addressing research gaps caused by the limitations of conventional experimental methods and providing deeper insights into how the δ phase affects the tensile properties of IN718 alloy.

2. Materials and methods

2.1. Heat treatment and tensile property test

The IN718 sample used in this study is a forged bar with a diameter of 100 mm, and its compositional details are presented in Table 1. The experimental samples were fabricated via electrical discharge machining (EDM) wire cutting. The heat treatment process directly affects the precipitation behavior of δ phase [30]. It is generally accepted that the precipitation temperature of the δ phase ranges from 780 °C to 900 °C [31], although some sources indicate a range of 930 °C–950 °C [32]. The precipitation rate of δ phase is maximized at 900 °C, with dissolution commencing at 980 °C and complete dissolution occurring between 1020 °C and 1038 °C. Additionally, the IN718 alloy exhibits a static recrystallization temperature of 1020 °C [6], and grain size remains unchanged when the heat treatment temperature is below 1020 °C. Consequently, a solution and aging heat treatment procedure was employed in this study. Initially, the IN718 alloy samples were heated to 1050 °C at a rate of 10 °C/min, held at this temperature for 1 h to dissolve the originally precipitated phases within the matrix, and then water-quenched to room temperature. Subsequently, the solution-treated samples were subjected to aging at temperatures ranging from 850 °C to 975 °C, with 25 °C intervals, for holding times of 1, 3, 5, 8, 12, and 24 h. The detailed heat treatment schedules are illustrated in Fig. 1.

To obtain microstructural information of the δ phase, the heat-treated samples were sequentially ground using 240#, 600#, 1200#, 3000#, and 5000# grit SiC papers until the surfaces were free from visible scratches. Subsequently, the samples underwent electrolytic polishing using a solution composed of perchloric acid (HClO₄) and anhydrous ethanol (C₂H₆O) at a 1:9 (v/v) ratio. The polishing was

performed at a constant voltage of 18 V for 15 s. After polishing, the microstructure on the sample surfaces became clearly distinguishable. To ensure model generalizability, microstructural images of the IN718 alloy were acquired using two scanning electron microscopes (SEMs) with different imaging qualities: CIQTEK SEM 5000 and Tescan S8000. SEM images were captured at a of 140 × magnification with resolutions of 3072 × 2048, 6144 × 4096, and 8448 × 5632 pixels, including two images acquired per condition.

To evaluate the tensile properties of the IN718 alloy under various heat treatment conditions, specimens were machined into standard dog-bone-shaped specimens with gauge dimensions of 15 mm × 3.2 mm × 1.5 mm, as shown in Fig. 2. Tensile tests at room temperature were conducted using an E45.205 universal testing machine (Yingst Corporation) under strain-controlled loading at a rate of $5 \times 10^{-3} \text{ s}^{-1}$. The tensile property values for each condition were obtained as the average of three parallel specimens.

2.2. Construction of the δ phase segmentation model

2.2.1. Experiment dataset

To enable intelligent quantification of the δ phase in IN718 alloy, a dedicated dataset for δ phase identification and segmentation was first constructed in this study. To ensure the model's adaptability to variations in δ phase precipitation states and imaging resolutions, SEM images of IN718 samples aged at 900 °C for 1, 5, 8, and 12h were selected. Each condition was represented by two images with resolutions of 3072 × 2048, 6144 × 4096, and 8448 × 5632 pixels. A total of 1184 image patches were manually annotated using Labelme software. After excluding those without δ phase, 1019 labeled image pairs were retained. The annotations were then converted into binary masks, in which pixels corresponding to the δ phase were assigned a value of 1, and all background pixels were assigned a value of 0. The dataset preparation workflow is illustrated in Fig. 3.

All 1019 labeled samples were subsequently split into a training set (611 images), validation set (204 images), and test set (204 images) in a ratio of 6:2:2.

2.2.2. Model construction

This study employed a deep learning model termed TernaNet-16, which integrates the VGG-16 [33] encoder with the U-Net [34] decoder to segment the δ phase in the microstructure of IN718 alloy. The model aims to maximize the feature extraction proficiency of VGG-16 and the spatial segmentation precision of U-Net for semantic segmentation tasks. The TernaNet-16 architecture comprises two main components: the encoder and the decoder, linked by skip connections to preserve spatial information across multiple scales, as illustrated in Fig. 4. The encoder is based on the VGG-16 model. To reduce training expenses and enhance model performance, it employs a transfer learning approach [35]. Initially, it is pre-trained using the ImageNet dataset [36], which offers a robust deep convolutional backbone for feature extraction. The decoder is engineered to reconstruct the segmented output while reducing overfitting via transposed convolution and dropout regularization. The encoder contains five convolutional blocks, each with padding to maintain spatial dimensions and max-pooling operations that downsample feature maps by a factor of two, enabling hierarchical feature capture from fine to coarse structures—features closely related to δ phase morphology. The decoder mirrors the encoder

Table 1
Chemical composition of IN718 alloy.

Element	C	Si	Mn	S	P	Cr	Fe	Ni
wt%	0.032	0.12	0.11	0.002	0.005	18.46	balance	52.11
Element	Mo	Al	Nb	Ti	Co	Cu	Mg	B
wt%	2.95	0.44	5.02	0.99	0.26	0.016	0.002	0.003

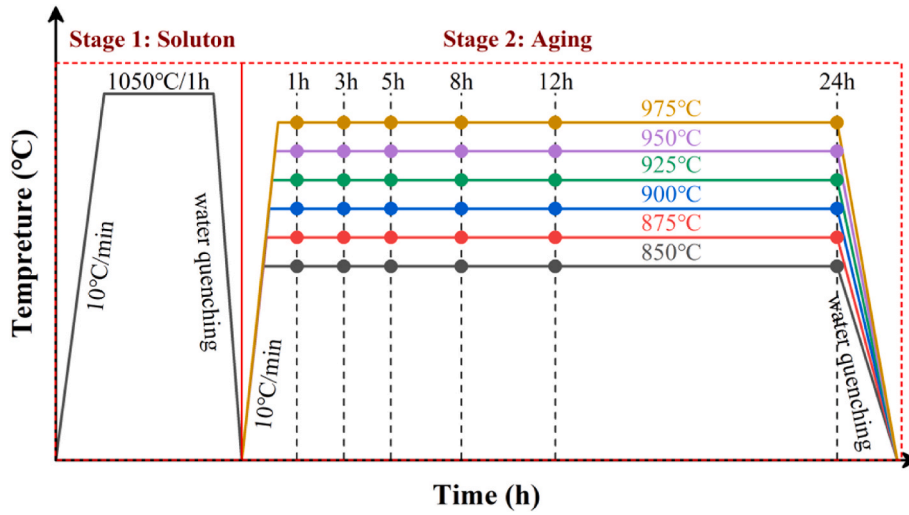


Fig. 1. Solution and aging heat treatment of IN718 alloy.

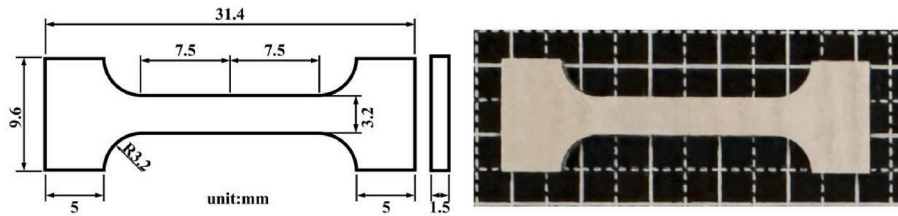


Fig. 2. Dimensions of tensile specimen.

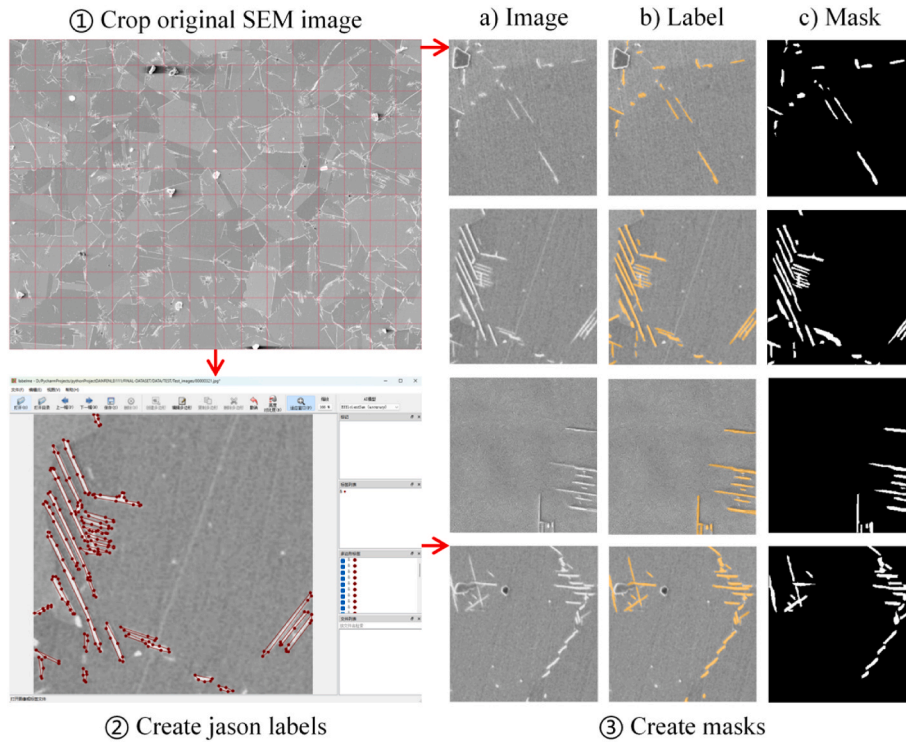


Fig. 3. Workflow of making δ phase segmentation model dataset.

structure and performs upsampling to restore spatial resolution. Each decoding block includes ReLU activation, dropout (rate = 0.5), and transposed convolution layers for effective feature reconstruction. The

final output layer employs a 1×1 convolution to convert the 32-channel feature map into a single-channel prediction mask, enabling pixel-wise δ phase classification.

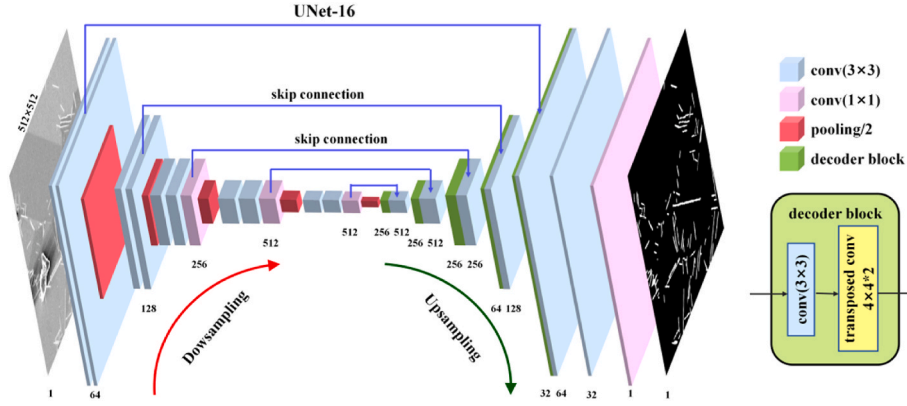


Fig. 4. TernausNet-16 network structure.

2.2.3. Model hyperparameters

To enhance the model's accuracy, we conducted data augmentation on the initial training set photos, employing techniques such as rotation, flipping, scaling, and Gaussian blurring, resulting in an expanded training set of 1596 images. The learning rate (lr) was set to 5e-4, the batch size at 4, the number of training epochs at 200, and the regression network was optimized using the Adam algorithm. A ReduceLROnPlateau scheduler was applied to automatically diminish the learning rate by a factor of 0.5 if the validation loss failed to improve across 5 consecutive epochs, thereby helping prevent overfitting and accelerating convergence. The network was trained and tested on a single machine equipped with an Intel(R) Core(TM) i9-12900 K CPU @ 3.19 GHz, GeForce RTX 3090, 64 GB RAM, running Windows 11 - 64-bit, Python 3.10, and PyTorch, employing data parallelism to enhance model training efficiency.

The loss function quantifies the disparity between predicted and true values, serving as a feedback mechanism to optimize parameters. This study employed a composite loss function to enhance the model's segmentation performance. This loss function primarily comprises Cross-Entropy Loss and Dice Loss (designated as $Loss_{ce}$ and $Loss_{dice}$, respectively), which balances classification accuracy and target region consistency through a weighted summation defined by the following formula:

$$Loss_{ce} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log (1 - \hat{y}_i)] \quad (1)$$

$$Loss_{dice} = 1 - \frac{\sum_i y_i \hat{y}_i + \epsilon}{\sum_i y_i + \sum_i \hat{y}_i + \epsilon} \quad (2)$$

$$Loss = \alpha Loss_{ce} + \beta Loss_{dice} \quad (3)$$

where N denotes the batch size (set to 4 in our implementation), y_i is the ground-truth label of sample i , p_i is the original output of the model, $\hat{y}_i$ is the probability value after Sigmoid normalization, ϵ is the smoothing term (set to 1e-5), α and β are the weighting coefficients (both set to 1), which are adjusted according to the trial-and-error method on the basis of the validation set.

Given the sparsity of δ phase under some conditions and the strong class imbalance, segmentation performance was assessed using Dice Coefficient, Precision, Recall, and Intersection over Union (IoU), calculated as follows:

$$Dice = \frac{2TP}{2TP + FP + FN} \quad (4)$$

$$IoU = \frac{TP}{TP + FN} \quad (5)$$

$$Precision = \frac{TP}{TP + FN} \quad (6)$$

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

where TP (true positives) refers to the number of samples correctly predicted as belonging to the positive class; FP (false positives) represents the number of instances incorrectly predicted as positive; FN (false negatives) indicates the number of positive instances misclassified as negative; and TN (true negatives) refers to the number of instances correctly predicted as negative.

2.3. Construction of the tensile performance prediction model

2.3.1. Original dataset construction and data processing

In the tensile performance prediction module, the δ phase segmentation model was initially applied to extract δ phase features from 216 cropped SEM images corresponding to 36 distinct sample types. The resulting prediction masks were used to aggregate quantitative data of δ phase. To characterize the geometry of the segmented δ phases, we employed a minimum bounding rotated rectangle approach, which fits the smallest possible rectangle with arbitrary orientation that completely encloses each δ phase particle. This method allows for an accurate measurement of the major axis length of each δ phase. For each sample type, measurements from six SEM images were averaged to calculate statistical values. The statistical results included the following characteristics: the total area percentage of δ phase (Total AP), the average δ phase length (Total AL), the area percentage of needle-like δ phase (Needle AP), the area percentage of rod-like δ phase (Rod AP), and the area percentage of granular δ phase (Granular AP), which were selected as part of model input features. Building on this foundation, the original dataset was constructed using tensile strength (UTS), yield strength (YS) and elongation (EL) as model output features. The construction of the final dataset is detailed in Section 3.3.1.

Due to the inherent characteristics of material experiments, the collection of experimental data was typically laborious, and the number of samples was relatively small. To improve model performance and reduce the risk of overfitting due to limited data, we augmented the original dataset using the Synthetic Minority Oversampling Technique (SMOTE) from Python's imbalanced-learn module. SMOTE is a prevalent technique in machine learning that synthesizes new samples inside the feature space, effectively augmenting data diversity while circumventing the information redundancy associated with mere repetitive sampling [37]. The technique operates by selecting a sample and interpolating between it and one of its five nearest neighbors (determined by Euclidean distance) to synthesize a new data point. The target sample size was established at 300. For each original sample x_i , the

algorithm identifies its five nearest neighbors based on Euclidean distance within the feature space, randomly selects one neighbor x_m , and subsequently interpolates between the two to create a new sample using a formula: $x_{new} = x_i + \lambda \bullet (x_m - x_i)$ (where λ is a random number in $[0,1]$). Concurrently, the same interpolation method, $y_{new} = y_i + \lambda \bullet (y_m - y_i)$ is applied to the output variables to generate the corresponding values. Through the iterative use of this procedure, the total sample size was augmented to 300, thus providing sufficient and diverse data to support the following training of the predictive model. Furthermore, to mitigate discrepancies among varying feature scales, the input features and target outputs are standardized using the Z-score method, which transforms the original data to have zero mean and unit variance. The standardization formula is as follows:

$$x_{std} = \frac{x - \mu}{\sigma} \quad (8)$$

where x is the raw data, μ is the mean, σ is the standard deviation, and x_{std} is the standardized data.

2.3.2. Models for predicting tensile performance

Eight regression algorithms were employed for model training in this study, including Gaussian Process Regression (GPR) [38], Random Forest Regression (RFR) [39], Support Vector Regression (SVR) [40], Bayesian Linear Regression (BLR) [41], Gradient Boosted Regression (GBR) [42], Deep Neural Networks (DNN) [43], Kernel Ridge Regression (KRR) [44], and Multilayer Perceptron (MLP) [45]. To reduce the risk of overfitting in nonlinear regression models, a 5-fold cross-validation strategy was adopted, with the dataset split into 80 % training and 20 % testing. All models underwent hyperparameter optimization to ensure optimal performance. Specifically, a grid search method was conducted to generate a comprehensive set of hyperparameter combinations, and suboptimal configurations were eliminated based on validation performance. Model evaluation was conducted using two standard metrics: the Root Mean Square Error (RMSE) and the coefficient of determination (R^2), which are defined as follows:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (9)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y}_i)^2} \quad (10)$$

where N is the number of datasets, y_i is the measured value, $\hat{y}_i$ is the predicted value of the dependent variable of the regression model, and $\bar{y}_i$ is the mean value of the dependent variable.

3. Results and discussion

3.1. Microstructure and mechanical properties analysis

3.1.1. Analysis of the precipitation behavior of δ phase

The microstructures of IN718 alloy under 36 different heat treatment conditions was characterized using SEM, as shown in Fig. 5. The precipitation behavior of the δ phase varied significantly depending on the heat treatment parameters. In the temperature range of 850 °C–975 °C, δ phase content increased with extended aging time. At 850 °C, δ phase precipitation exhibited low sensitivity to aging time, and the overall quantity of precipitation remained minimal. This phenomenon can be attributed to the fact that 850 °C lies near the intersection of the precipitation curves of the γ' and δ phases. At this temperature, the precipitation rate was generally low, and the γ' phase competes with the δ phase for Nb atoms, thereby inhibiting δ phase precipitation [46].

Conversely, δ phase precipitation progressively increases with rising temperature after the aging temperature exceeds 850 °C. Simultaneously, as the temperature rose, the morphology of the δ phase transitioned from granular to rod-like, as illustrated in Fig. 6(a); with an additional temperature increase, the δ phase progressively transformed into needle-like morphology, as shown in Fig. 6(b). Furthermore, with prolonged aging time, the amount of needle-like δ phase also increased. In addition, the intragranular δ phase was observed to align in parallel clusters, whereas intergranular δ phases primarily appeared as isolated particles near grain boundaries. These precipitation patterns are consistent with those reported in the literature [6,47,48].

3.1.2. Mechanical property analysis

Fig. 7 illustrates the stress–strain curves of 36 sample groups subjected to different aging treatments, while Table 2 summarizes the

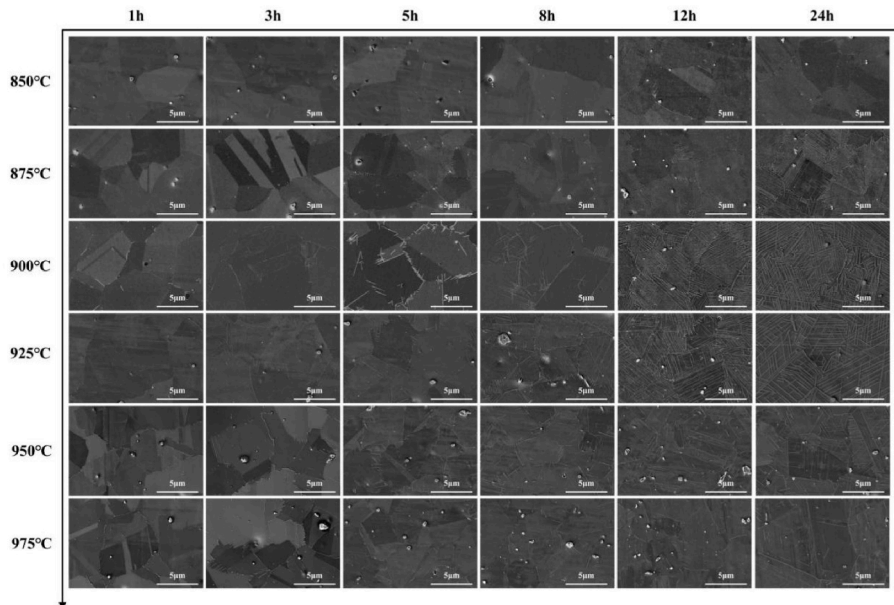


Fig. 5. Microstructure images of IN 718 alloy under various aging treatments.

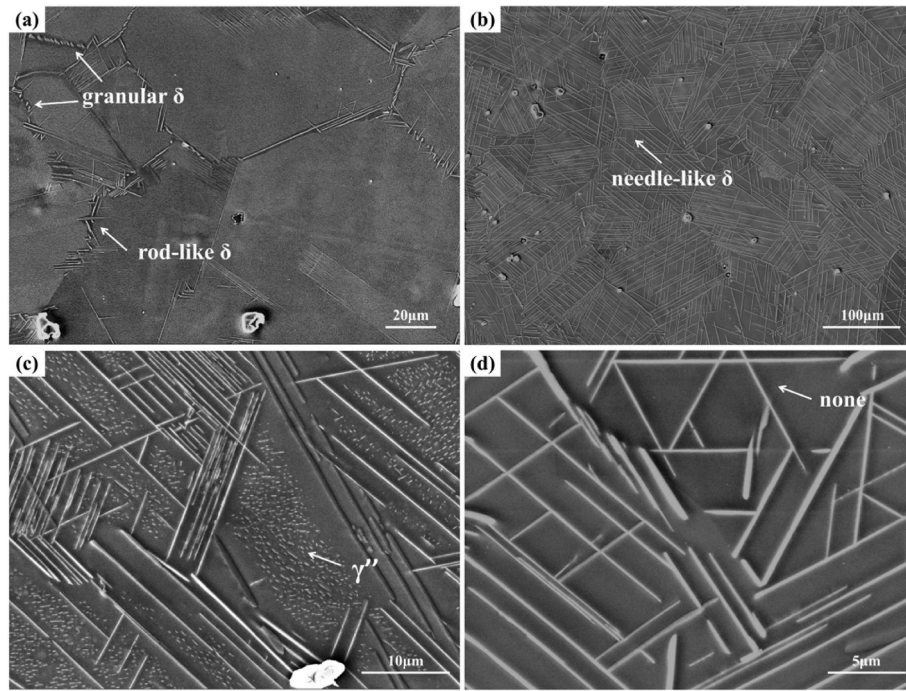


Fig. 6. (a, b) δ phase with varied morphologies, (c) intragranular γ' phase observed at aging temperatures below 925 °C, (d) absence of intragranular γ' phase at aging temperatures above 925 °C.

corresponding UTS, YS, and EL. The results indicate that UTS and YS reach their maximum values at an aging temperature of 850 °C for 1 h, while EL peaks at 975 °C for 5h. However, UTS, YS, and EL did not show discernible linear or nonlinear correlations with aging temperature and time, necessitating further analysis of the influence of these microstructural features on tensile properties.

3.2. δ phase segmentation models

3.2.1. Model training and evaluation

We conducted an analysis of the TernaNet-16 model in comparison to the conventional U-Net, the TernaNet-11 model (with VGG-11 as the encoder), and the LinkNet-34 model (with ResNet-34 as the encoder). As the number of training epochs increased, the training and validation loss of TernaNet-16 decreased from 1.663 to 0.529 to 0.022 and 0.016, respectively. The other models showed similar downward trends and convergence behavior, indicating adequate training. Model performance was quantitatively evaluated using the four metrics introduced in Section 2.3.3, and the results are provided in Table 3. Among all models, TernaNet-16 achieved the highest evaluation metrics, with Dice, Precision, Recall, and IoU reaching 75.1 %, 73.2 %, 81.2 %, and 80.6 %, respectively.

Fig. 8 illustrates the segmentation results of the four models applied to the same dataset. In Case 1, TernaNet-16 and LinkNet-34 effectively avoid misclassification caused by carbide interference and appropriately identify the δ phase, but U-Net and TernaNet-11 erroneously classify some carbide regions as δ phase. In Case 2, a low-quality, noisy SEM image was used to assess model robustness. TernaNet-16 showed greater robustness to noise and successfully identified all δ phase regions, while LinkNet-34 showed the weakest performance, missing most δ phase features. Case 3 demonstrates the model's precision in segmenting the closely spaced δ phases. TernaNet-16 could effectively separate adjacent δ phases. However, the other models exhibited subpar performance in edge segmentation and are prone to contour merging issues between two adjacent δ phases.

Consequently, in terms of segmentation accuracy, TernaNet-16 outperforms the other three models, demonstrating proficient δ phase

detection and segmentation capabilities, thereby establishing a crucial basis for the ensuing statistical analysis of δ phases.

To evaluate the generalization capability and practical applicability of the segmentation model, we assessed its performance across multiple datasets acquired from three distinct SEMs with varying resolutions, as detailed in Table 4. The results indicate that the TernaNet-16 model achieved Dice and IoU scores below 70 % on images with a resolution of 2048×1536 pixels obtained from the FEI Helios NanoLab 650 SEM. Except for this case, the model demonstrates consistently significant segmentation performance across the other image sets. These findings suggest that TernaNet-16 is relatively robust to variations in SEM imaging conditions and exhibits competent segmentation performance even at 4K resolution. Therefore, the model offers strong generalizability and practical value, reducing reliance on high-quality SEM images and alleviating the burden on materials researchers.

Subsequently, we evaluated the segmentation accuracy of the model under several typical heat treatment conditions. Three representative settings were selected: 850 °C for 8h, 925 °C for 5h, and 925 °C for 12h. These conditions respectively represent low δ phase precipitation, varied δ phase morphology, and widespread δ phase distribution. As shown in Fig. 9(a–f), the TernaNet-16 model demonstrated excellent adaptability to changes in δ phase characteristics, maintaining high sensitivity and accuracy in diverse scenarios.

Ultimately, we examine the space and time complexity of each model, which are critical metrics for assessing the model's practical applicability. Fig. 9(g) summarizes the model capacity and temporal complexity of several techniques. The model's capacity is quantified by the number of parameters requiring training, represented by blue bars; the time complexity is indicated by the number of 512×512 -pixel metallographic images processed per second (unit: FPS). Among the evaluated models, the basic U-Net network has the smallest parameter count at 7.9M. TernaNet-16, with a parameter count of 32.3 million, slightly exceeds the other models but provides significantly better segmentation accuracy. LinkNet-34 exhibits the highest prediction speed, at 95.29 FPS, making it appropriate for segmenting extensive datasets when accuracy requirements are not stringent. Although slower than LinkNet-34, TernaNet-16 processes 512×512 -pixel images at 33.13

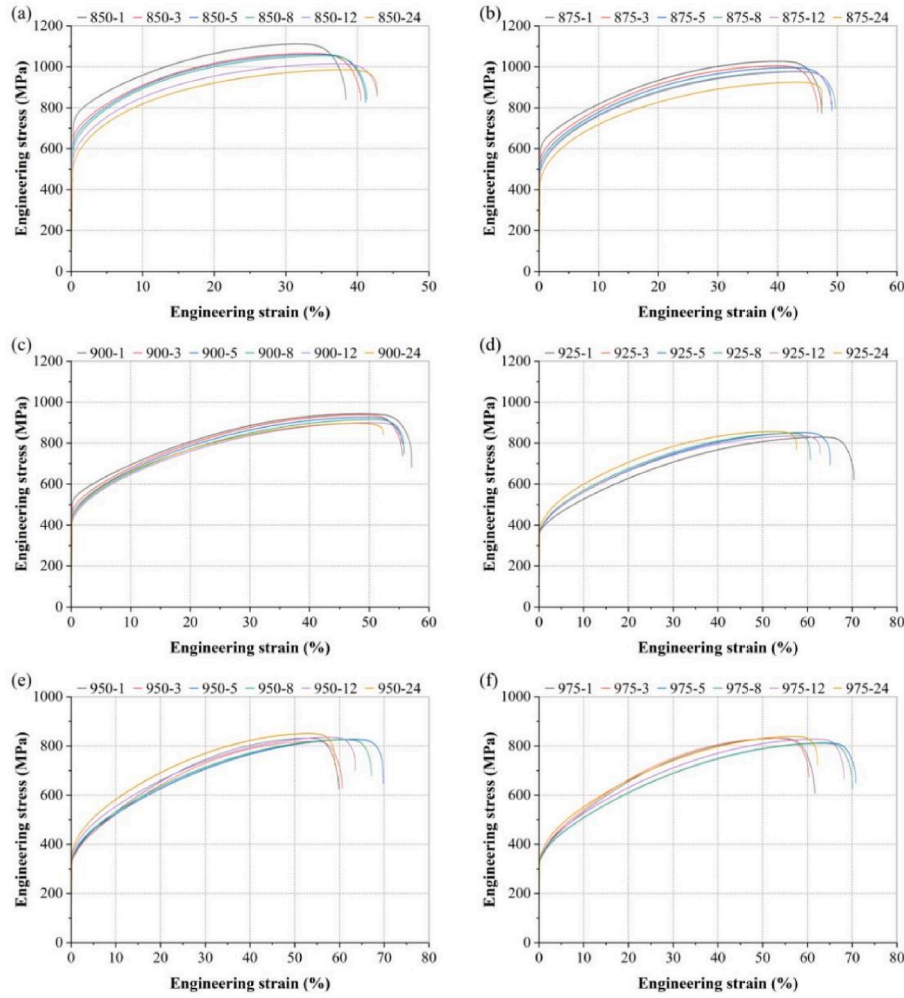


Fig. 7. Stress-strain curves of IN718 alloy at aging temperatures of 850–975 °C with different holding times.

Table 2

UTS, YS, and EL of IN718 alloy under different aging treatments.

	UTS/MPa					
	850 °C	875 °C	900 °C	925 °C	950 °C	975 °C
1h	1114	1029	946	831	819	831
3h	1067	1007	940	832	831	830
5h	1062	998	927	852	829	814
8h	1056	980	916	850	827	813
12h	1017	978	901	836	837	829
24h	988	926	898	858	852	840
	YS/MPa					
	850 °C	875 °C	900 °C	925 °C	950 °C	975 °C
1h	1069	938	810	631	644	648
3h	1021	918	801	638	647	656
5h	1014	902	790	670	629	613
8h	1004	884	774	679	637	612
12h	958	880	763	668	662	633
24h	924	831	773	710	695	663
	Elongation/%					
	850 °C	875 °C	900 °C	925 °C	950 °C	975 °C
1h	38.37	47.42	57.11	70.39	60.65	61.68
3h	40.49	46.73	55.59	69.80	59.83	60.35
5h	41.16	49.12	55.85	65.11	69.77	70.81
8h	41.35	49.70	55.78	60.68	67.14	70.10
12h	42.73	48.98	55.34	62.82	63.54	68.24
24h	42.79	47.45	52.33	57.57	58.94	62.24

Table 3

Evaluation metrics for the four segmentation models.

Models	Dice/%	IoU/%	Precision/%	Recall/%
U-Net	71.69	58.26	71.39	76.93
TernausNet-11	74.40	61.70	72.74	80.26
TernausNet-16	75.06	80.57	73.18	81.19
LinkNet-34	74.72	61.80	73.13	80.82

FPS, and is capable of segmenting a 4096×6144 -pixel image in under 3 s.

Given TernausNet-16's notable superiority in segmentation precision and generalization capacity, it was selected for subsequent statistical analysis of the δ phase.

3.3. Mechanical property prediction model

3.3.1. Final dataset construction

We applied TernausNet-16 to segment the δ phase in SEM images of 36 sets of heat-treated samples, followed by quantitative analysis. For δ phase morphology quantification, we employed the minimum rotated rectangle method to extract geometric data from each closed contour. Phases with a longest side $\leq 2 \mu\text{m}$ were classified as granular, those $> 2 \mu\text{m}$ and $\leq 8 \mu\text{m}$ as rod-like, and those $> 8 \mu\text{m}$ as needle-like. The overall results of content and morphology quantification are shown in Fig. 10. The most rapid δ phase precipitation occurred within the temperature range of 900 °C–925 °C.

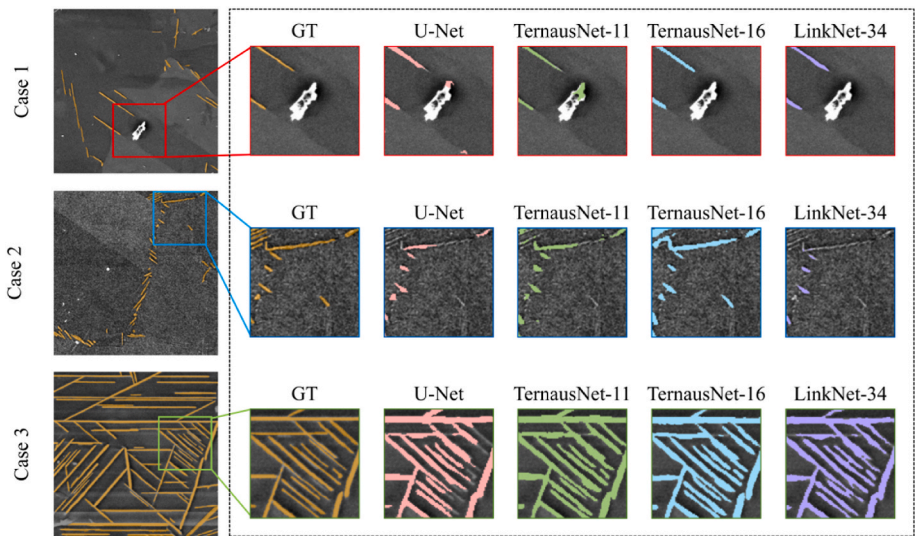


Fig. 8. Segmentation results of the four models in different cases.
Note: GT stands for Ground Truth.

Table 4
Dice and IoU of TernaNet-16 on images of different resolutions acquired by different SEMs.

	FEI Helios NanoLab 650		CIQTEK SEM 5000		TESCAN S8000		
Resolution/px	2048*1536	4096*3072	3072*2048	6144*4096	8448*5632	4096*3072	8192*6144
Dice/%	65.96	74.30	71.85	72.21	75.54	73.64	80.62
IoU/%	68.40	76.0	73.20	75.73	79.70	79.16	81.48

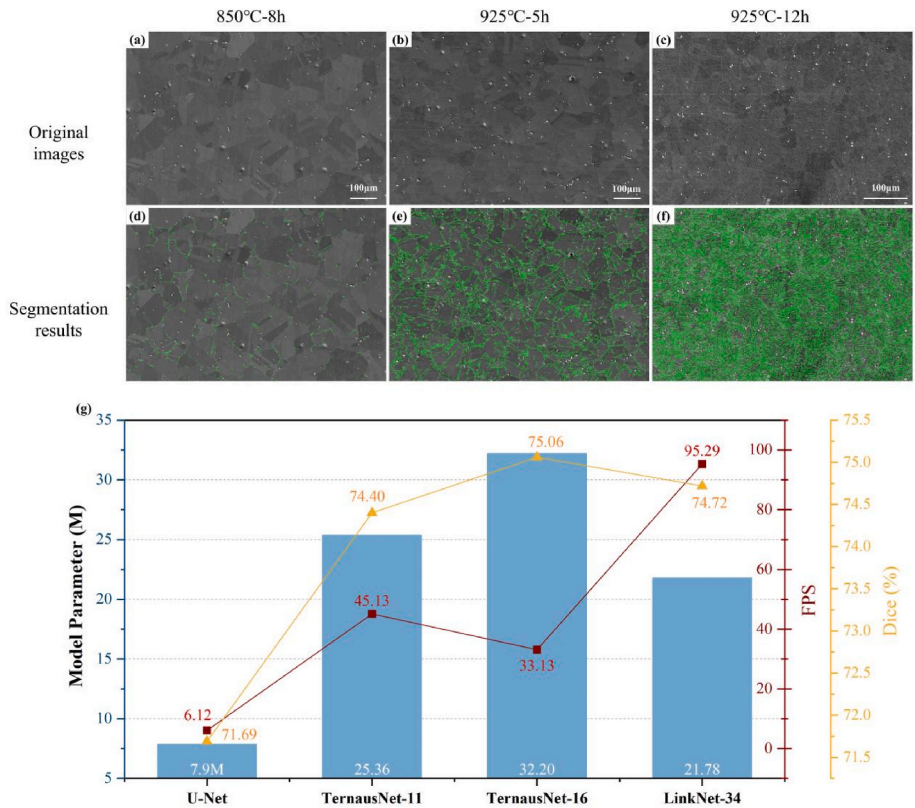


Fig. 9. (a–c) original images and (d–f) segmentation results of TernaNet-16 at 850°C-8h, 925°C-5h, and 925°C-12h; (g) Capacity, FPS and Dice for the four models.

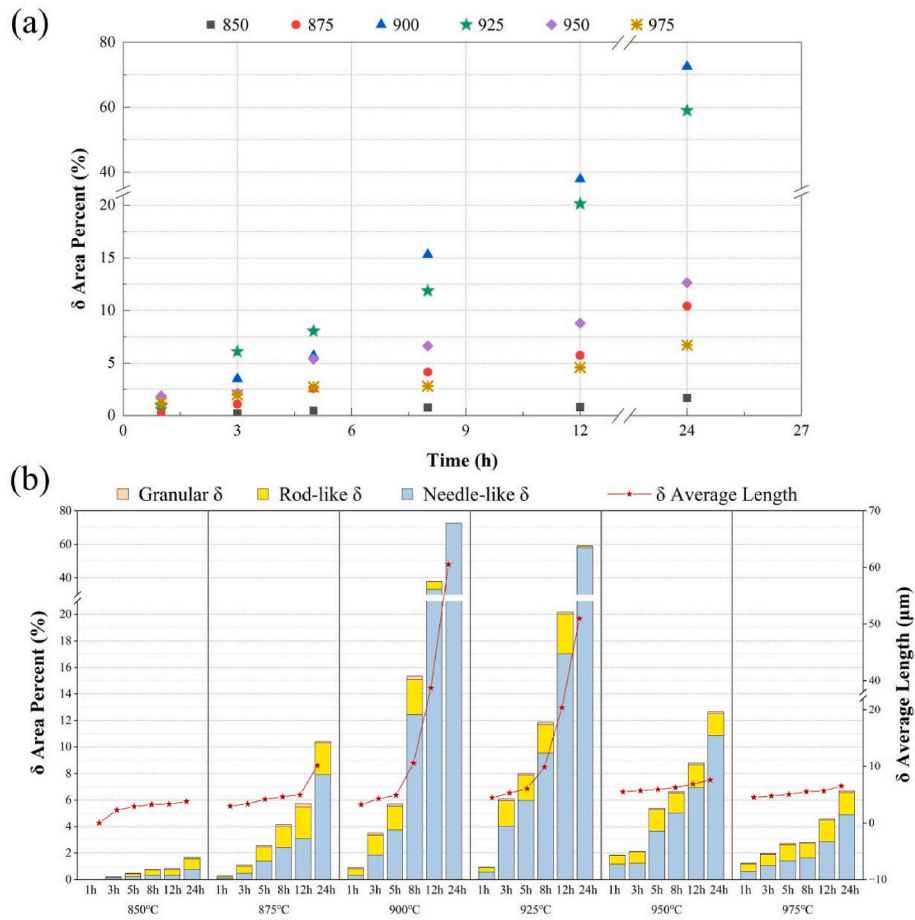


Fig. 10. (a) Total area percentage of the δ phase under various aging treatments; (b) Area percentages and average lengths of needle-like, rod-like and granular δ phases under various aging treatments.

IN718 alloy is a precipitation-hardening superalloy, composed of the γ matrix phase, the primary strengthening phase γ'' , the secondary strengthening phase γ' , δ phase, carbides, and other phases. Among them, the γ'' phase plays a crucial role in determining the alloy's mechanical properties. Its precipitation temperature ranges from 595 °C to 870 °C, with peak precipitation rates observed between 732 °C and 760 °C, and its dissolution temperature lies between 870 °C and 930 °C [49]. However, when aged at temperatures above 800 °C, the precipitation of the δ phase consumes some of the Nb, thereby reducing the amount of γ'' phase and diminishing the alloy's precipitation-strengthening effect. We observed the presence of granular γ'' phase in intragranular regions at aging temperatures below 925 °C, as shown in Fig. 6(c and d), with an average size of approximately 0.15 μm . In contrast, γ' is a secondary strengthening phase, with a precipitation temperature range of 593 °C–816 °C and a dissolution temperature range of 843 °C–871 °C [50], which was not observed in this study.

Therefore, the γ'' phase content was incorporated into the input parameters of the prediction model to enhance the accuracy of performance predictions. According to the precipitation law of the γ'' phase, its content is defined as follows: when the temperature is ≤ 925 °C, the content of the γ'' phase is given by the formula: $\gamma'' = \frac{1}{\text{Time} \times \text{Temperature}}$, and when the temperature exceeds 925 °C, the content of the γ'' phase content was set to zero. Simultaneously, grain size analysis reveals that, under a fixed solution treatment temperature, variations in aging temperature and duration in this work have a negligible effect on grain size among the 36 sample sets. Thus, the "microstructure-tensile property" dataset was finally developed utilizing γ'' , Total AL, Needle AP, Rod AP, and Granular AP as inputs, with UTS, YS, and EL as outputs.

3.3.2. Efficacy of predictive models

Fig. 11(a) shows the prediction performance of various regression models. Among them, the RFR model achieved the best performance, yielding the lowest RMSE and highest R^2 values for UTS, YS, and EL. As shown in Fig. 11(b), the predicted values closely align with the experimental results, with most data points lying close to the diagonal line, indicating strong predictive accuracy.

RFR is an ensemble learning method that constructs multiple decision trees using bootstrapped samples and randomly selected feature subsets. This strategy introduces diversity among trees, reduces model variance, and mitigates overfitting [39]. Predictions are then averaged across all trees to produce the final output, resulting in a robust model capable of capturing complex nonlinear relationships. In this study, RFR demonstrated superior performance due to its intrinsic advantages, especially in handling small datasets with high-dimensional and nonlinear input features. Unlike traditional linear models, RFR does not require complex feature transformations or kernel design. Moreover, it is relatively insensitive to hyperparameter tuning and provides reasonable interpretability when combined with SHAP analysis. Considering its strong predictive performance, robustness, and interpretability, RFR was selected as the primary model for subsequent SHAP-based analysis to investigate the influence of δ phase features on tensile properties.

3.3.3. Interpretability analysis

Feature relevance refers to each feature's relative contribution to model output, essential for interpretation. After training the model on the dataset, we evaluated the importance of each feature using SHAP. Fig. 12(a–c) presents the SHAP value visualizations for UTS, YS, and EL. The swarm plots illustrate the distribution of SHAP values for each

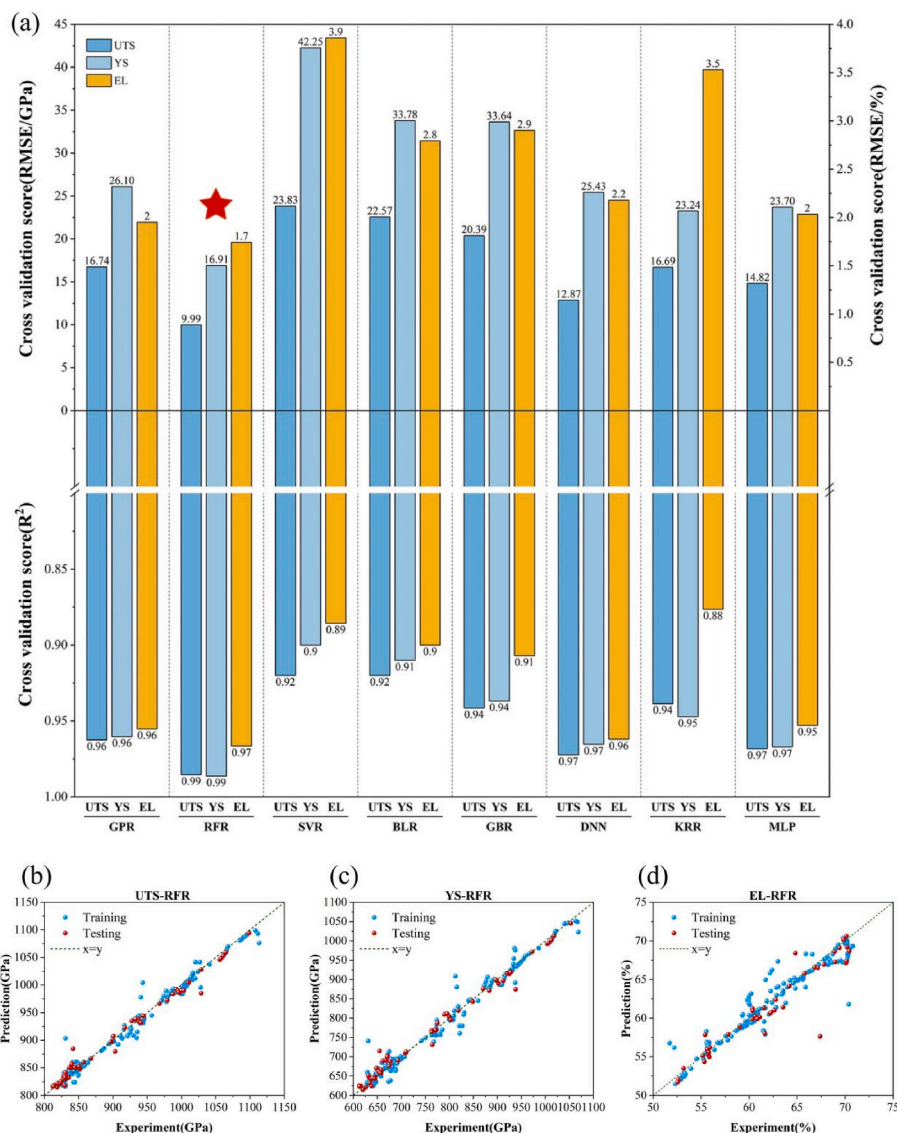


Fig. 11. (a) RMSE and R^2 for eight regression models; (b) RFR-based prediction results for UTS, YS, and EL.

feature, where the color denotes the magnitude of the feature value, and the position indicates its positive or negative contribution to the prediction. The corresponding light blue bar plots quantify the global importance of each feature.

Additionally, existing studies indicate that the δ phase's influence on tensile properties is not definitely beneficial or detrimental. An optimal δ phase quantity can significantly enhance mechanical performance, whereas excessive amounts may impair it. To decouple the individual and interactive effects of δ and γ'' phases, we employed an integrated approach combining SHAP (SHapley Additive exPlanations) dependence plots with Generalized Additive Models (GAM). This dual methodology provides three key advantages: (1) SHAP values quantify the isolated contribution of each phase by calculating marginal effects across all possible phase combinations; (2) GAM captures nonlinear responses through smooth functions while explicitly modeling phase interactions via interaction terms [22]; (3) The combination allows visualization of both individual contributions (SHAP dependence plots) and their synergistic effects (GAM partial dependence plots). Based on feature ranking, the top three contributors were selected for further analysis, as shown in Fig. 13(a–i). The results reveal that among all outputs, the γ'' phase is the most influential factor, indicating its dominant role in affecting the tensile behavior of IN718 alloy. This

finding is consistent with prior studies by Li and Rizea et al. [49,51], which demonstrated that a fine dispersion of γ'' phase promotes lattice distortion and significantly improves tensile strength. The SHAP dependence plots indicate that the γ'' phase positively contributes to both UTS and YS, but has a negative effect on EL.

For UTS, the second most influential feature is Total AL, indicating that the δ phase also exerts a considerable impact on it. Analysis of Fig. 13(d) indicates that Total AL exerts a beneficial effect on UTS when it is less than 6.03 μm , during which the δ phase precipitates at the grain boundaries as granular, rod-like, and partially needle-like forms, while there is negligible δ phase precipitation in intragranular regions, where the δ phase precipitation has minimal impact on the γ'' phase.

A reduction in UTS was noted when Total AL was within the range of 6.03–30.37 μm . This may result from competition between δ and γ'' phases for Nb atoms. Within this precipitation interval, increased δ phase precipitation at elevated aging temperatures consumes Nb, thereby suppressing γ'' formation. Additionally, a number of δ phases are transformed directly from the γ'' phase, thereby diminishing its presence. This can be further supported by the SHAP interaction plot between γ'' and Total AL in Fig. 13(j). The different colors represent the SHAP value under the interaction of the two features. The dark blue section on the left side of the image indicates that when γ'' approaches 0,

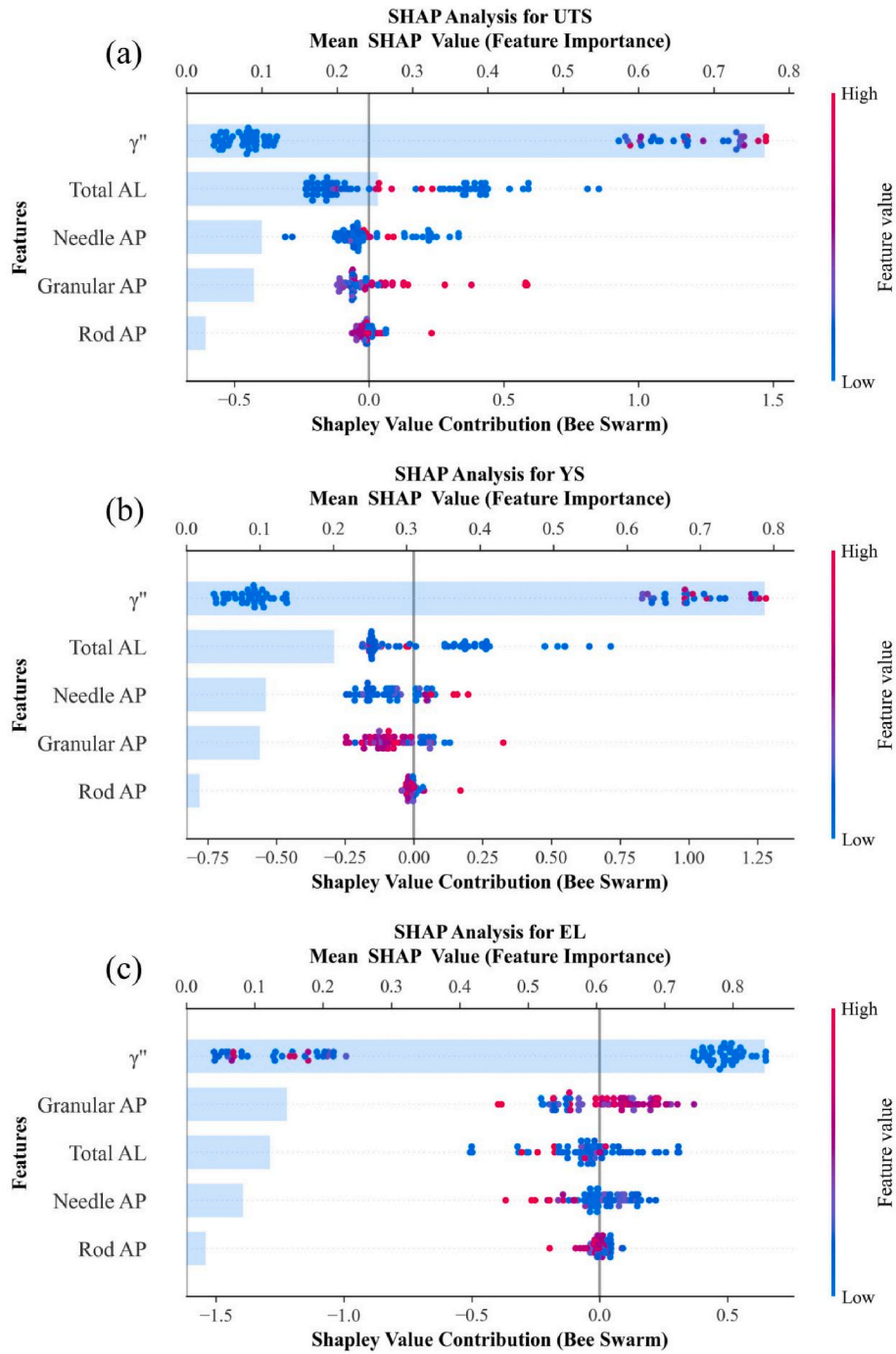


Fig. 12. Contribution of each input feature to (a) UTS; (b) YS; and (c) EL.

the interaction effect between γ'' and Total AL on UTS is pronounced, resulting in a negative impact of Total AL on UTS within this range.

When Total AL exceeds 30.37 μm , UTS begins to increase again. This phenomenon was also observed when Needle AP surpasses 25.16 %, which is noteworthy. A similar trend was also found in the influence of the δ phase on YS. Specifically, YS exhibited a positive response when Total AL was below 5.82 μm or above 31.4 μm , as well as when Needle AP was below 2.66 % or above 16.15 %. Previous literature often categorizes δ phase influence as either beneficial at low concentrations or detrimental when excessive. Considering that γ'' begins to dissolve at temperatures above 870 $^{\circ}\text{C}$ and that the extensive precipitation of δ phase consumes Nb, thereby reducing the formation of γ'' . This change in γ'' state would only lead to a decrease in UTS and YS. Therefore, the

influence of γ'' was ruled out. To further investigate this behavior, we analyzed the precipitation strengthening mechanism.

The strengthening mechanisms of IN718 alloy primarily include solid-solution strengthening, grain boundary strengthening, dislocation strengthening, and precipitation strengthening, with the contributions of these mechanisms being either independent or additive. The contribution of precipitation strengthening can be estimated using the Orowan equation:

$$\sigma_p = \frac{0.4M\mu b \ln\left(\frac{d}{b}\right)}{\pi\bar{\lambda}\sqrt{1-\nu}} \quad (11)$$

where M is the Taylor factor (3.06) [52], μ is the shear modulus (79

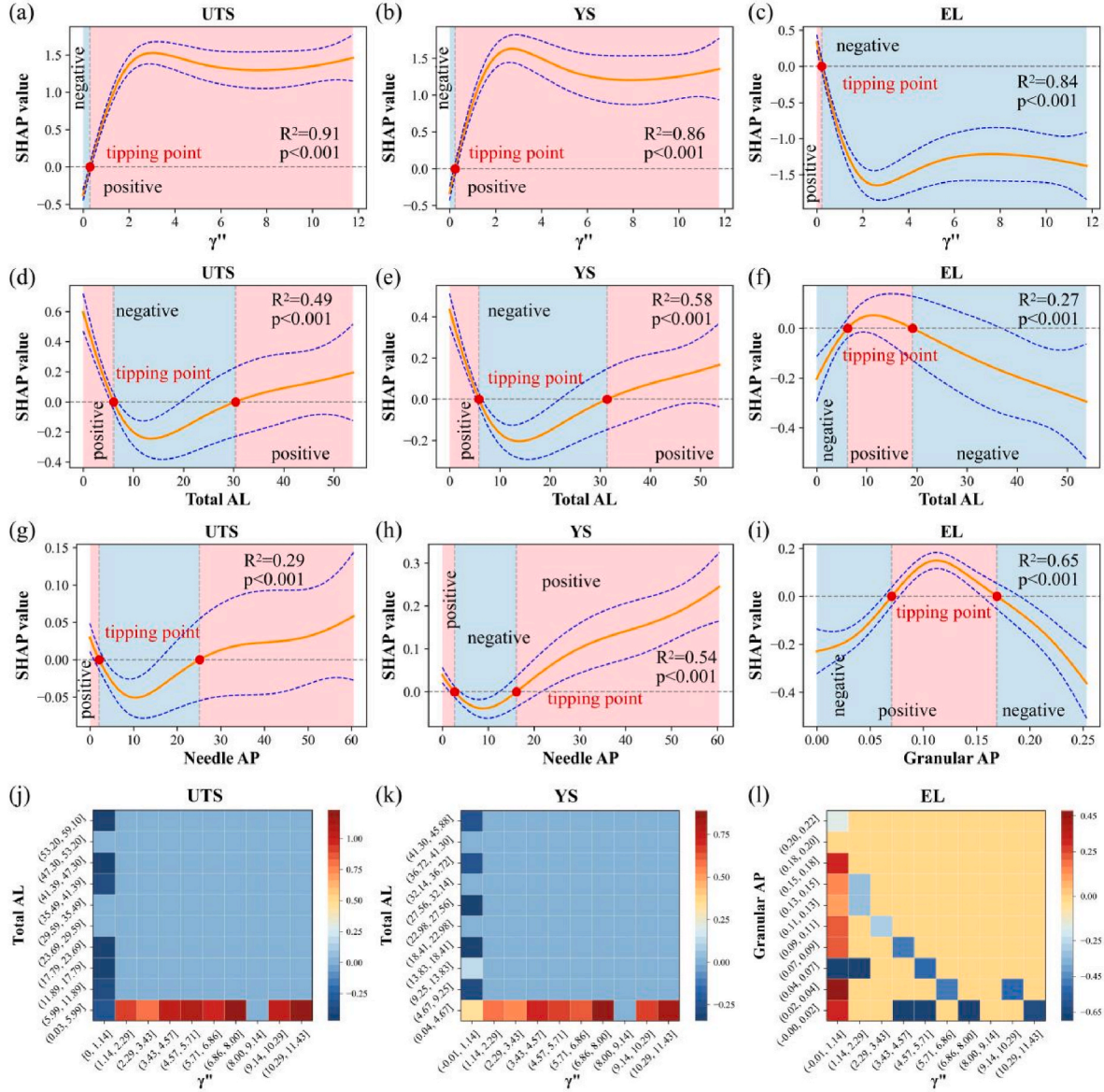


Fig. 13. (a–i) SHAP dependence plots for each input feature of the UTS, YS and EL prediction models; heat maps showing the interaction effects of (j, k) γ'' and Total AL on UTS and YS; and (l) γ'' and Granular AP heatmap of the interaction effect on EL.

GPa), b is the Burgers vector (0.2495 nm) [53], σ_p represents the contribution from precipitation strengthening, $\bar{d}$ and $\bar{\lambda}$ are the average diameter and average spacing of the precipitated phases, respectively, and ν is Poisson's ratio (0.35).

We evaluated the strengthening contribution under aging conditions of 900 °C and 925 °C for 12h and 24h, which were the most apparent cases of UTS and YS enhancement. Quantitative analysis of δ phase features using TernaNet-16 revealed that at 900 °C, the average precipitate diameter $\bar{d}$ increased from 38.72 μm after 12h to 60.51 μm after 24h, while the inter-particle spacing $\bar{\lambda}$ decreased from 10.10 μm to 1.85 μm , resulting in a marked increase in precipitation strengthening from 35.43 MPa to 200.76 MPa. A similar trend was observed at 925 °C, where $\bar{d}$ increased from 20.37 μm after 12h to 50.93 μm after 24h, and $\bar{\lambda}$ decreased from 6.35 μm to 3.18 μm , leading to an increase in precipitation strengthening from 53.29 MPa to 115.22 MPa. These findings explain the observed improvements in UTS and YS. Longer aging promotes both the precipitation and growth of δ phase particles, leading to an increase in δ phase density and a reduction in average inter-particle spacing. Furthermore, the interaction of nearby dislocations amplifies

the strengthening via dislocation entanglement. The trends of Total AL and Needle AP on YS mirror those for UTS, as shown in Fig. 13(e–h), and are not elaborated further here.

For EL, the two most influential features besides γ'' phase are Granular AP and Total AL. As shown in Fig. 13(f), Total AL values below 6.14 μm negatively impacted EL, likely due to grain boundary δ phases obstructing dislocation motion and thus hindering plastic deformation. Within the range of 6.14–19.12 μm , Total AL had a positive correlation with EL. This is attributed to the reduction of γ'' phase—whose decrease reduces alloy strength but increases ductility—outweighing the effects of δ phase precipitation. However, when Total AL exceeds 19.12 μm , EL decreases again, likely due to Orowan-type strengthening that impedes plastic deformation. Additionally, excessive needle-like δ phase precipitating within grains can induce microcrack initiation, further reducing ductility [41].

A beneficial influence of granular δ phase on EL was observed in the range of 0.05 %–0.17 %, as illustrated in Fig. 13(i). SHAP interaction analysis between Granular AP and γ'' , shown in Fig. 13(l), reveals a significant interactive region (highlighted in red) indicating a complex

interplay. However, this positive effect primarily stems from the reduced γ'' content; thus, when the γ'' phase's influence is excluded, granular δ generally exerts a negative effect on elongation.

4. Conclusion

This study addresses the lack of comprehensive research on the influence of the δ phase on the tensile properties of IN718 alloy, a gap limited by traditional methods. Leveraging machine learning, we developed a systematic approach to analyze this relationship. A TernaNet-16 model—based on U-Net and a pre-trained VGG-16 encoder—was trained on SEM images from three aging conditions. The model achieved strong performance (Dice: 75.1 %, Precision: 73.2 %, Recall: 81.2 %, IoU: 80.6 %) and showed robust segmentation across noisy and multi-resolution images.

Using the trained model, we quantified δ phase features in 36 aging-treated samples. These features, combined with γ'' phase data, were input into a Random Forest Regression model to predict tensile properties, achieving over 97 % accuracy (R^2). SHAP analysis further clarified the contribution of each microstructural feature. Key findings include:

- 1) δ phase content increases with aging time between 850 and 975 °C, peaking at 900–925 °C. At 850 °C, competition with γ'' limits δ phase growth.
- 2) The δ phase enhances UTS via two mechanisms: boundary pinning when its average length is above 6.03 μm , and precipitation strengthening when the total length and needle-like fraction exceed 30.37 μm and 25.16 %, respectively. A similar trend is also observed for YS. For EL, both a small amount of δ phase precipitated along grain boundaries and a dense distribution of δ phase exert negative influences.
- 3) While γ'' has a more dominant effect, the δ phase still plays a significant role. Its impact is nonlinear and depends on its precipitation characteristics and interaction with γ'' .

This work introduces a machine learning framework that effectively segments microstructures and predicts mechanical properties, overcoming data limitations in conventional studies. It demonstrates the potential of machine learning to advance microstructural analysis and property prediction in materials science.

CRediT authorship contribution statement

Zhixuan Cheng: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jianli Zhou:** Writing – review & editing, Supervision, Investigation, Data curation. **Guanghao Guo:** Writing – review & editing, Methodology, Formal analysis. **Shuang Chen:** Visualization, Resources, Investigation. **Xianjue Ye:** Writing – review & editing, Project administration, Data curation. **Xiao Wei:** Data curation, Conceptualization. **Yuefei Zhang:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Data curation, Conceptualization. **Ze Zhang:** Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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